Lecture 1 & 2: GIS and Data Modelling

GIS

- Provide tools for
  - Acquisition
  - Administration
  - Analysis
  - Visualisation

of **spatial Information**

Components:

- Hardware (sensors, computers)
- Software (programs, data)
  - leader in commercial sector: ArcGIS
  - free & open-source: QGIS

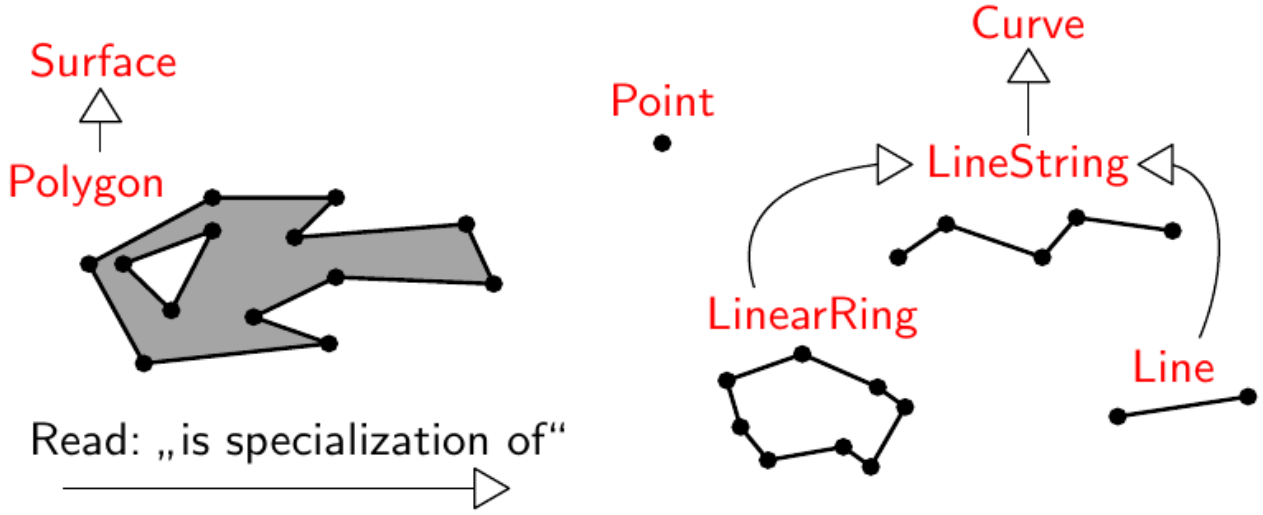
Raster and Vector Data

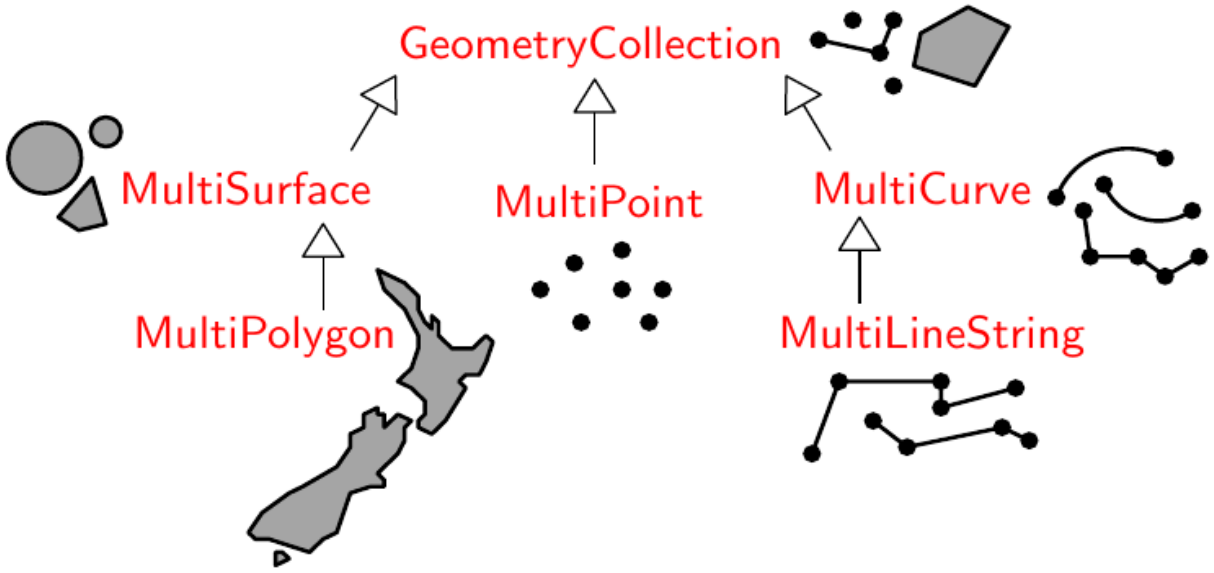
Raster Data

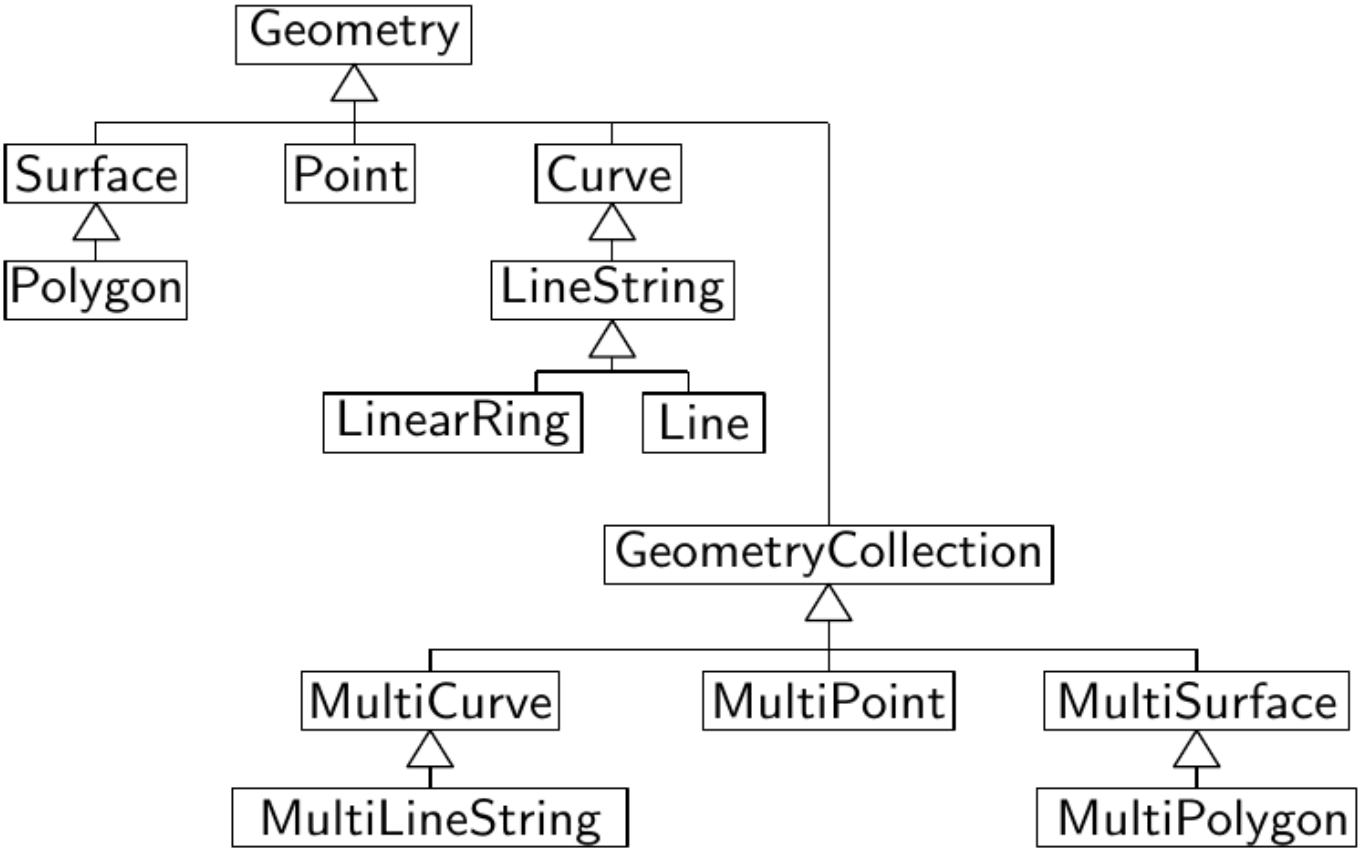
- Used for photos
- 3 channels (r, g, b) for each pixel
  - also other classes (e.g. land-cover class: forest)

Vector Data

- A **geometry** is represented with coordinates
- Geometry types are defined in GIS standard ISO 19125:

- 

  - other meaningful geometry types (not part of standard):
    - CircularArc (Specialisation of Curve)
    - Disk (Specialisation of Surface)
- Multi-Part Geometries:
  - 



WKT Format (Well Known Text)

- Representation of vector geometries as text
- Examples:
  - `POINT(30 10)`
  - `LINESTRING (30 10, 10 30, 40 40)`
  - `POLYGON ((30 10, 40 40, 20 40, 10 20, 30 10))`
  - `POLYGON ((35 10, 45 45, 15 40, 10 20, 35 10),(20 30, 35 35, 30 20, 20 30))`

Field-based and object-based view

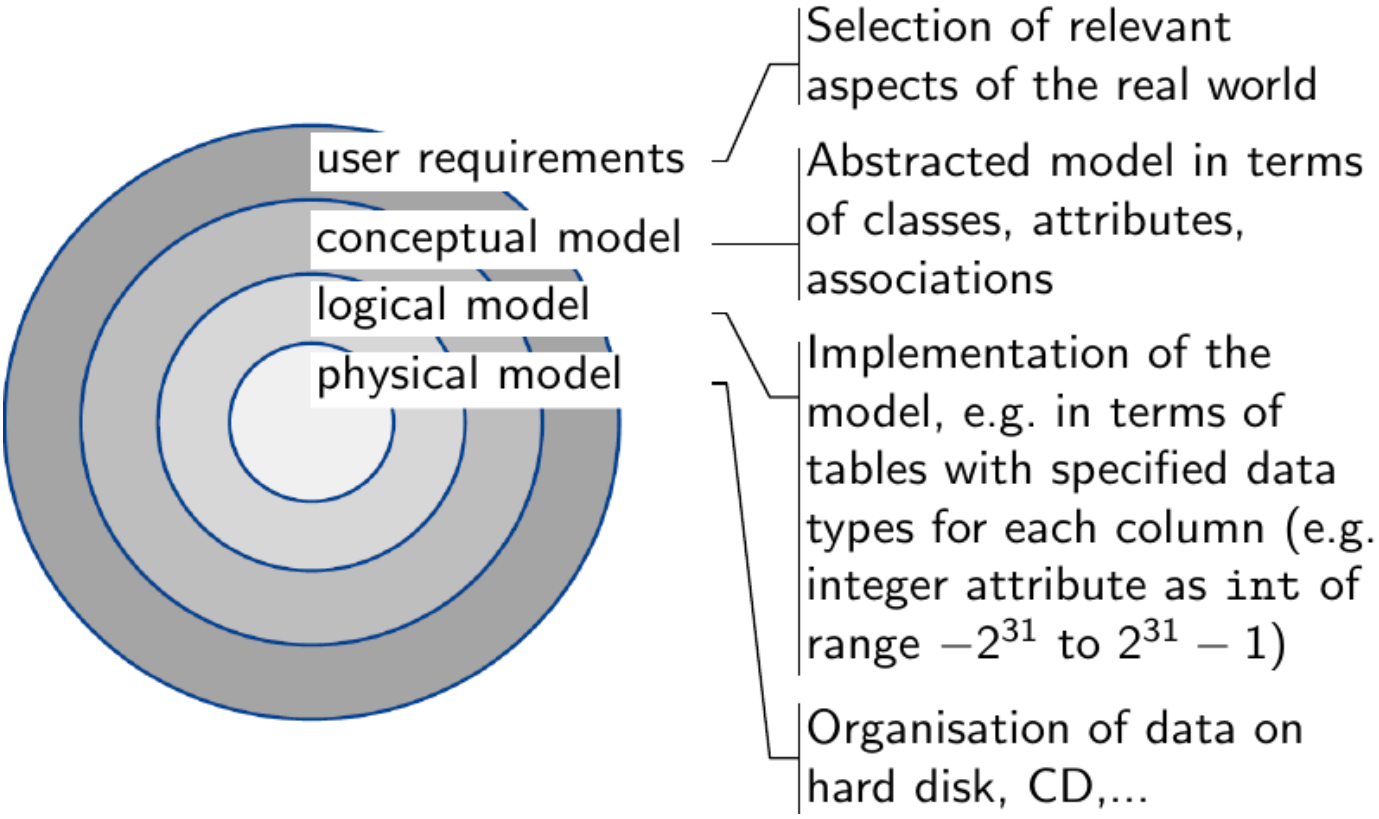
- Non-spatial attributes, i.e. **semantics**
- Approaches to model real world:
  - **Field**: Each point to value ($f : \mathbb{R}^2 \to \mathbb{R}$)
    - typically in physics: magnet field, gravity field
    - Geographical phenomena: surface height, land-cover type
    - Mathematical terms: function
      - continuous value field: ($f : \mathbb{R}^2 \to \mathbb{R}$)
      - discrete value field/ categorical coverage:($f : \mathbb{R}^2 \to \mathbb{Z}$)
    - With **raster**: pixels mapped to value from a range S
    - With **vector**: partitioned polygons mapped to value from range S

        - Polygons must not overlap and must not have gaps
        - Boundary between polygons of the same class is not meaningful
        - **All polygons constitue a polygonal subdivision**
    - With **data points**:
        - Set $P = \{p_1, ..., p_n\}$ of points with values $f(p_{1:n})$ given
        - For each other $p \in \mathbb{R}$ a value $f(p)$ can be computed according to **interpolation rule**
- **Objects**: Discrete real-world entities (buildings, streets) identified and modelled with shapes and attributes
  - *autonomous, unique* object
  - Consists of non-geometrical attributes and a geometry
  - Building A
  - Street X

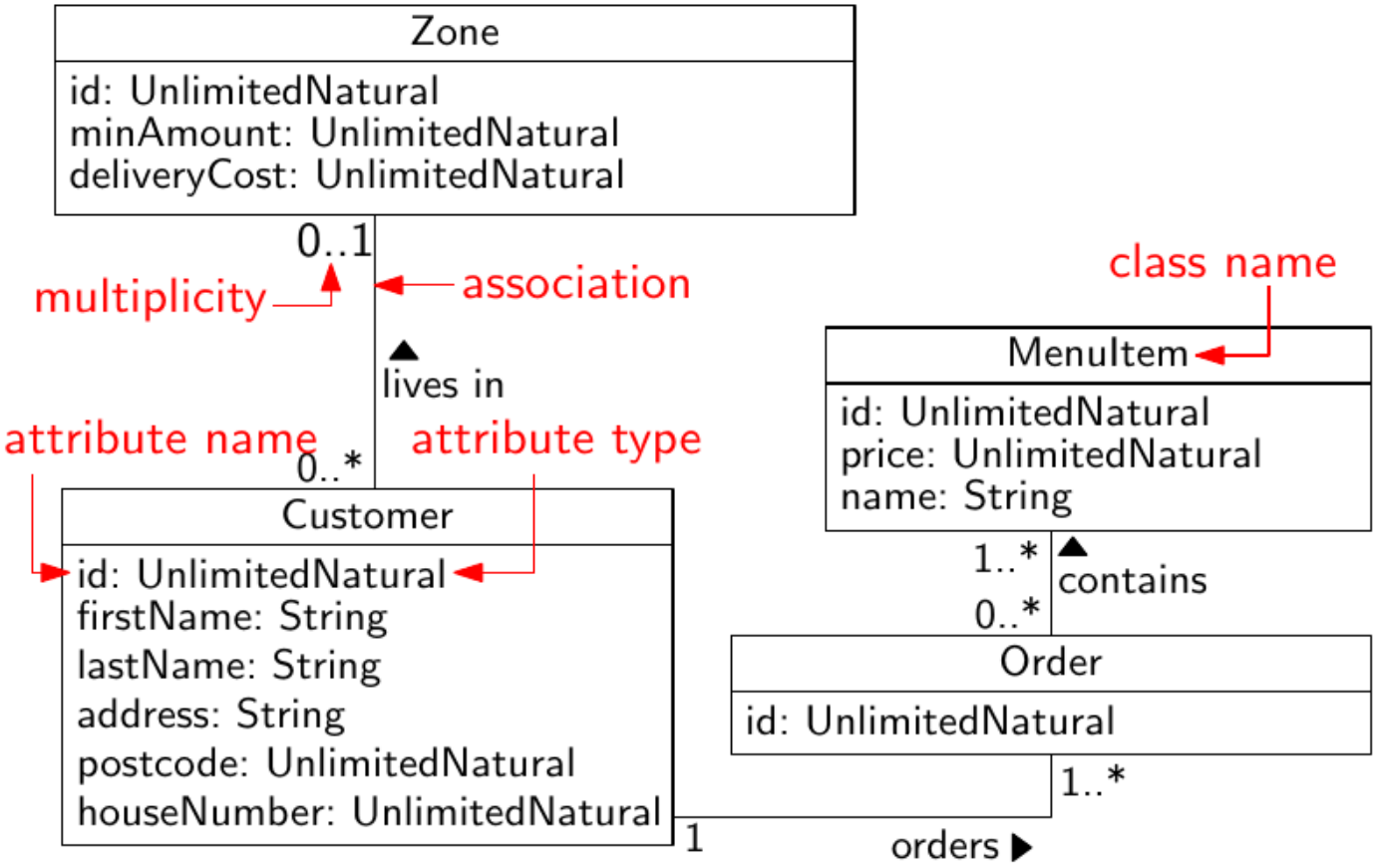
```
- Every object is member of class -> class|object|attribute
```

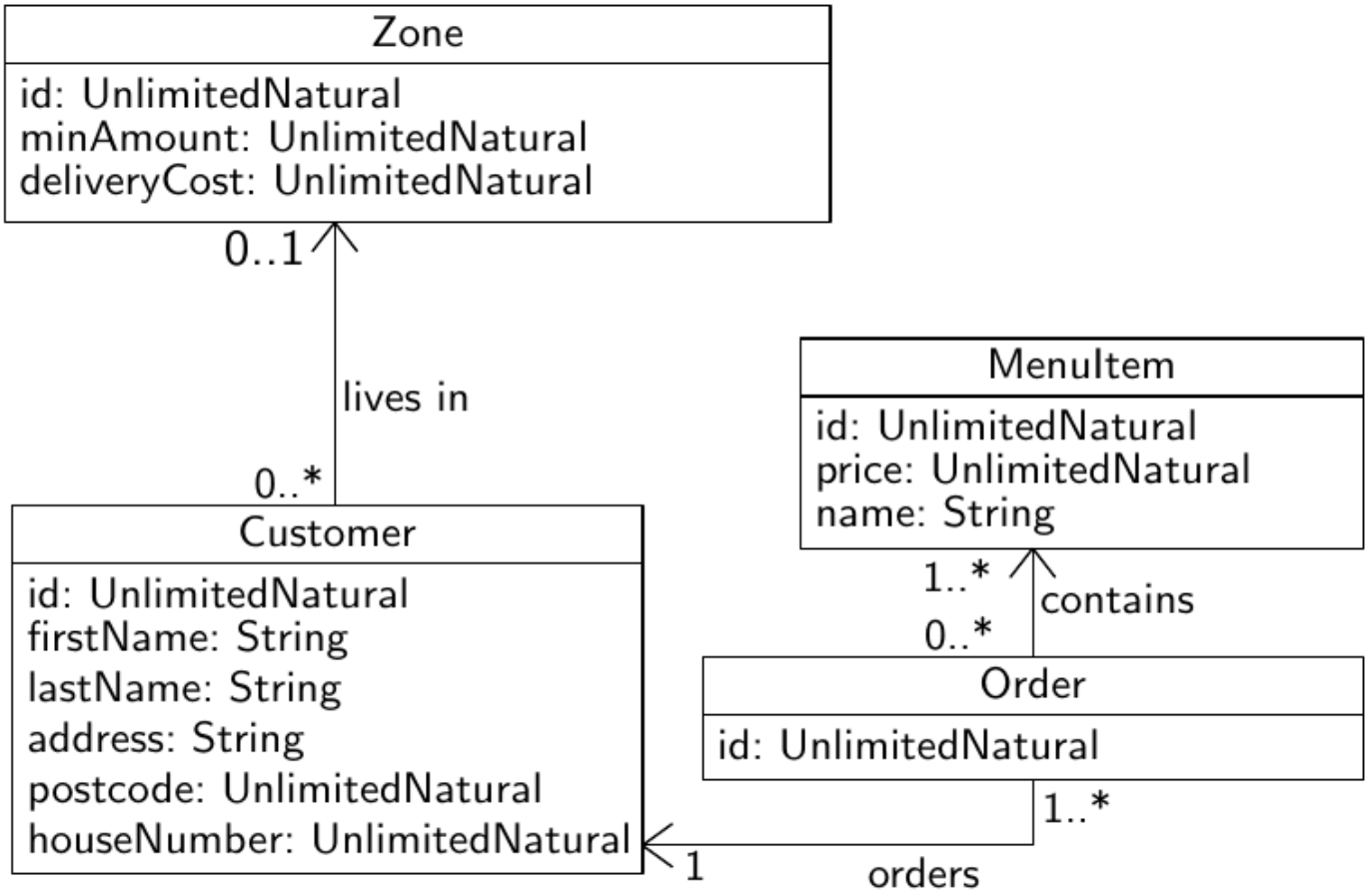
Data Modelling

- **Real-world entities** modelled as **objects** (polygons with attributes)
- **Modelling**: abstracting things and selecting certain aspects for a given purpose



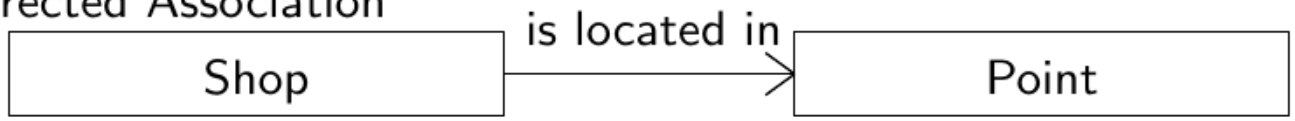
Conceptual Model with UML





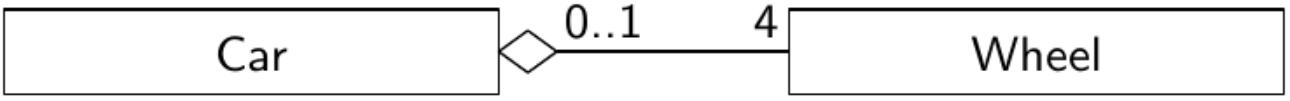
Special Associations

Directed Association



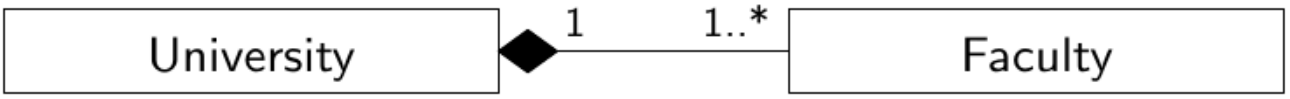
- navigable in only one direction (from Shop to Point)

Aggregation



- a special association meaning "is part of"

Composition



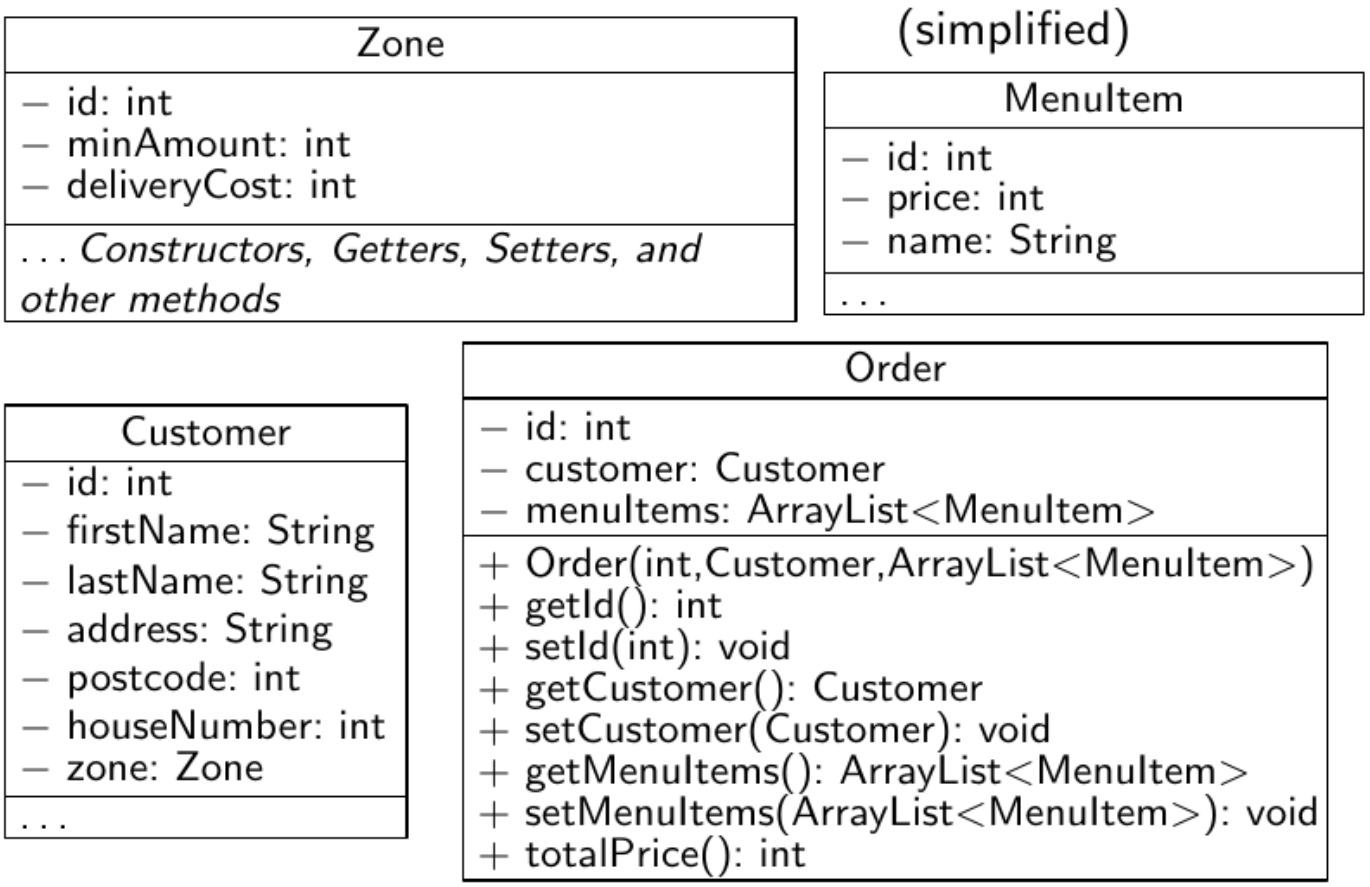
- a special association meaning "is part of" such that the parts (Faculties) cannot exist without the whole (University)

Two possible implementation of the conceptional model are possible:

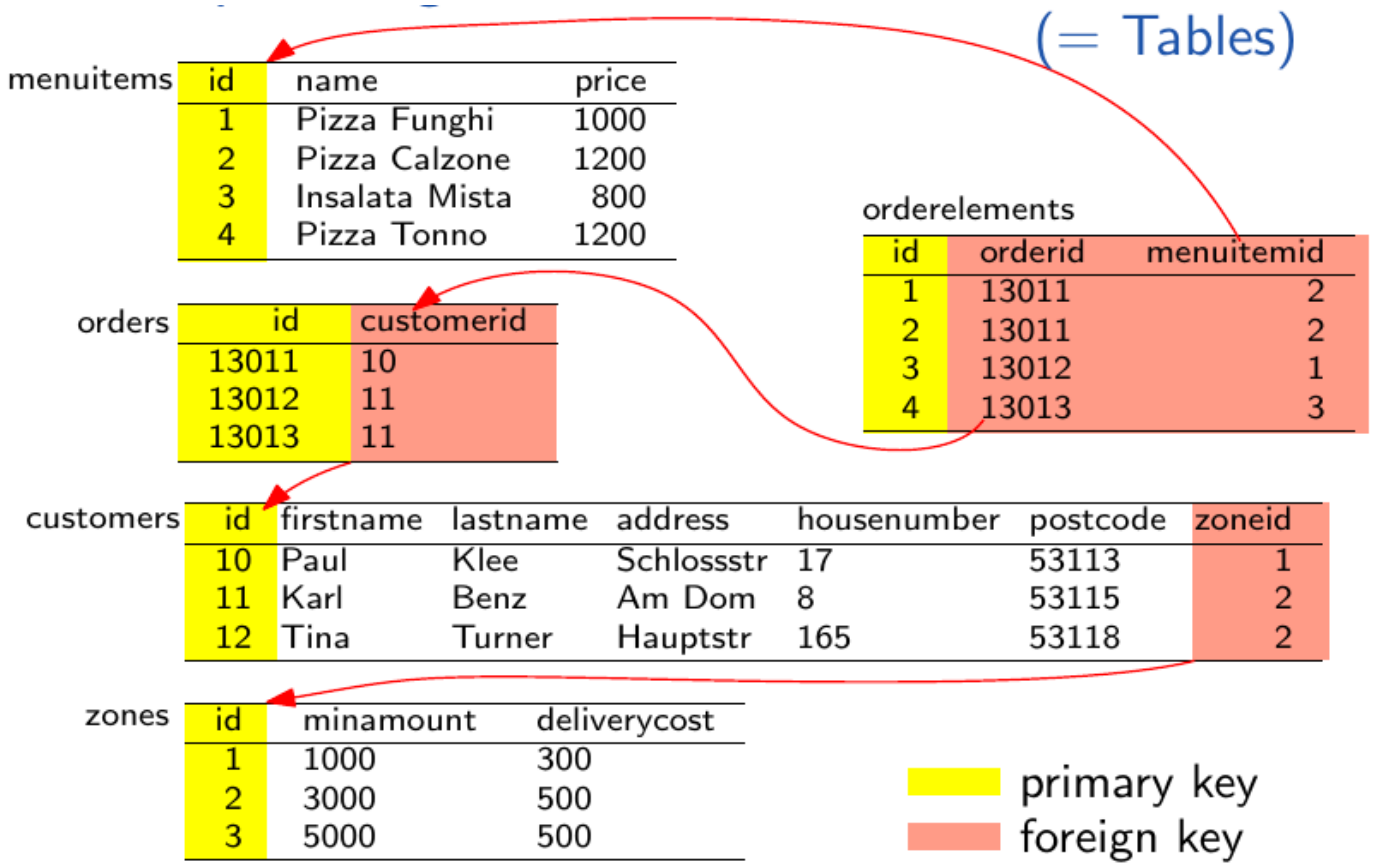
- classes of an object-oriented Java program
- tables of a relational database

Implementation of a Conceptual Model as a Java Program

Attribute types (Concpetual Model)	Java Data types (logical model)
unlimitedNatural, $\mathbb{N}$	int
Real, $\mathbb{R}$	double
String	String



Implementation of a Conceptual Model as a Database



Files vs. Databases

- **Separate Files**:

- One file (multiple files) for each table
- problem: **logical integrity** can be violated
  - i.e. data are logically correct and do not contradict:
    - e.g. Key that does not exist
    - concurrent usage

- **Databases**:
  - Interaction via database management system (DBMS)
  - Responsible for
    - Efficient execution of queries
    - Ensuring logical consistency after each transaction

Queries with SQL

WHERE

Which customers live in the street whose name is Schlossallee?

```
SELECT * FROM customers WHERE address = 'Schlossallee';
```

all columns — table — requirement for selected rows

ORDER BY

Return one the most expensive meals.

```
SELECT * FROM menuitems ORDER BY price DESC LIMIT 1;
```

descending order — only first row

Aggregate Functions

Return the highest prize among all meals.

```
SELECT MAX(price) FROM menuitems;
```

aggregate function

AVG()	average
COUNT()	total number
FIRST()	first value
LAST()	last value
MAX()	maximum
MIN()	minimum
SUM()	sum

GROUP BY

- only use with aggregate function or
- show only grouped column (at SELECT)

For each meal that has ever been ordered, return the id of the first order in which that meal occurs.

```
SELECT menuitemid, MIN(orderid) FROM orderelements
GROUP BY menuitemid;
```

minimum is computed separately for each group

NESTING

Return all the most expensive meals.

```
SELECT * FROM menuitems
WHERE price = (SELECT MAX(price) FROM menuitems);
```

SQL queries can be nested!